Mining High-Quality Pseudoinstance Soft Labels for Weakly Supervised Object Detection in Remote Sensing Images

Xiaoliang Qian, *Member, IEEE*, Yu Huo, Gong Cheng, *Member, IEEE*, Chenyang Gao, Xiwen Yao, *Member, IEEE*, and Wei Wang

***Abstract*— Weakly supervised object detection in remote sensing images (RSI) is still a challenge because of the lack of instance-level labels, and many existing methods have two problems. First, most of the existing methods usually mine the pseudoground-truth (PGT) instances solely relying on proposal class scores (PCSs). Actually, the reliability of PCS is not enough because of the bird's eye view imaging and large-scale chaotic background of RSIs, and the instances with high PCS incline to cover the discriminative region, rather than the whole object. Second, the existing methods assign a one-hot label to each instance, and the label of the PGT instance is copied to its neighbor instances, which induces the misclassification problem to some extent. Actually, the probability that the neighbor instances contain the object with the same category is smaller than the PGT instance. For the first problem, the proposal quality score (PQS) is proposed for mining high-quality PGT instances, which contain PCS and dual-context projection score (DCPS). The DCPS is calculated through semantic segmentation and is employed to measure the completeness that each proposal covers an object. For the second problem, a pseudosoft label assignment (PSLA) strategy is proposed to assign a more precise soft label for each instance, where the soft label is determined by the spatial distance between each instance and its nearest PGT instance. The ablation study validates the effectiveness of the PQS and PSLA. The comprehensive comparisons with other WSOD methods on three popular benchmarks show the excellent performance of our method.**

***Index Terms*— Dual-context projection score (DCPS), proposal quality score (PQS), pseudosoft label, remote sensing image (RSI), weakly supervised object detection (WSOD).**

I. Introduction

OBJECT detection in remote sensing images (RSIs) has great significance for semantic understanding of RSIs and can be applied to unmanned aerial vehicle-based RSI interpretation [1], military [2], [3], hazard prediction [4], scene classification [5], [6], [7], and so on. With the progress of deep learning techniques [8], [9], [10], [11], [12], the performance of fully supervised object detection (FSOD) [13], [14], [15], [16], [17], [18] in RSIs have made satisfactory development. However, the class and location labels of many instances are required for the training of FSOD models. Obviously, manually annotating the location labels of a large number of instances is time consuming and requires high labor costs. To reduce the heavy labeling costs, weakly supervised object detection (WSOD) attracted lots of interest in recent years, because only image-level labels were required.

At present, the existing methods solve the WSOD task through multiple instance learning (MIL) [19], [20], [21], [22], [23]. Specifically, the input image is considered a bag of object instances, and the image-level labels are used to supervise the instance classifier under the MIL framework. A path-breaking MIL-based WSOD model is weakly supervised deep detection networks (WSDDNs) [21], which treats the WSOD task as a multilabel classification problem. The online instance classifier refinement (OICR) network [24] is another milestone work, which added multiple ICR branches on the basis of WSDDN. The pseudoground-truth (PGT) instances used for training each ICR branch were mined according to the proposal class score (PCS) inferred from the last ICR branch. Furthermore, the neighbor instances of PGT instances were assigned the same labels as PGT instances, and the rest of the instances was assigned to the background category.

Most of the existing methods adopted the OICR network as the basic framework and made various innovations [25], [26], [27], [28], which significantly improved the performance of WSOD. However, two major problems still exist. The first problem is that many existing methods only employ the PCS to mine PGT instances for training multiple ICR branches; however, the reliability of PCS is not enough. As a matter of fact, the confusion between different categories in RSIs is more serious than in natural scene images, because of the bird's eye view imaging and the large-scale chaotic background of RSIs. Furthermore, as shown in Fig. 1(a), the instances with high PCS incline to cover the most discriminative region, rather than the whole object.

Manuscript received 21 October 2022; revised 10 February 2023 and 9 March 2023; accepted 10 April 2023. Date of publication 13 April 2023; date of current version 26 April 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62076223, Grant 62071388, and Grant 62136007; in part by the Key Science and Technology Program of Henan Province under Grant 232102211018; and the Key Research and Development Program of Shaanxi Province under Grant 2021ZDLGY01-08. *(Corresponding authors: Gong Cheng; Wei Wang.)*

Xiaoliang Qian, Yu Huo, and Wei Wang are with the College of Electrical and Information Engineering, Zhengzhou University of Light Industry, Zhengzhou 450002, China (e-mail: qxl_sunshine@163.com; yuhuo_henry2022@163.com; wangwei-zzuli@zzuli.edu.cn).

Gong Cheng, Chenyang Gao, and Xiwen Yao are with the School of Automation, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: gcheng@nwpu.edu.cn; cy_gao222@mail.nwpu.edu.cn; yaoxiwen@nwpu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3266838

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:06:51 UTC from IEEE Xplore. Restrictions apply.

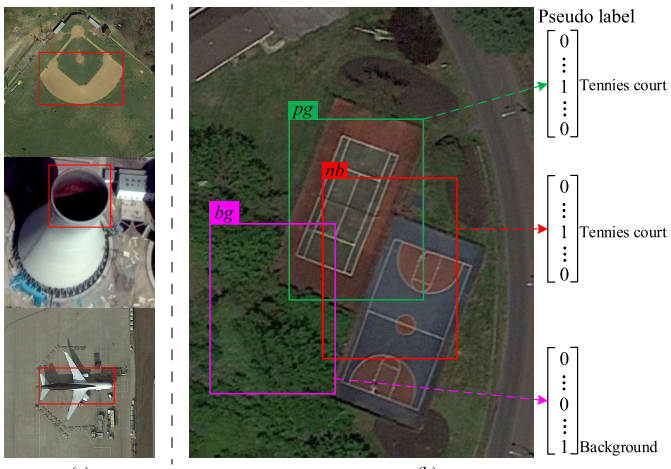


Fig. 1. Illustrations of the problems handled by our method. (a) Most discriminative regions, rather than the whole objects are usually detected because only PCS is used to mine PGT instances. (b) Misclassification problem induced by coping the label of PGT instance to its neighbor instances, where pg denotes PGT instance, nb denotes the neighbor instance whose IoU with pg is greater than or equal to 0.5, and bg denotes the background instance whose IoU with pg is less than 0.5. Note that pg is the nearest PGT instance of nb and bg. The same is below.

The second problem is that the neighbor instances of PGT instances were usually assigned the same label as their nearest PGT instance, which induces the misclassification problem to some extent. Actually, the probability that the neighbor instances contain the object with the same category is smaller than the one of the PGT instance; therefore, the neighbor instances should not be equally assigned the same label. Similarly, the labels of background instances should not also be assigned 1 in the background category, because the spatial distance between each background instance and its nearest PGT instance is different. For example, as shown in Fig. 1(b), the label of pg, i.e., tennis court, is copied to nb; however, the main content covered by nb is a basketball court. Obviously, the misclassification of nb will occur. For the same reason, the label of bg should not be assigned 1 in the background category because the tennis court is partially covered by bg.

In order to handle the above-mentioned problems, a novel pseudoinstance soft label mining (PISLM) module is proposed for WSOD in RSIs, as shown in Fig. 2. Specifically, to handle the first problem, the proposal quality score (PQS) is proposed and is adopted as the new metric to mine PGT instances. The PQS consists of two parts: the traditional PCS and the dual-context projection score (DCPS) of the proposal. The DCPS is calculated through weakly supervised semantic segmentation and is employed to measure the completeness that each proposal covers an object. Therefore, the PGT instances mined by PQS are more reliable and can better capture the whole object than the ones mined through PCS.

To handle the second problem, a novel pseudosoft label assignment (PSLA) strategy is proposed to assign more precise soft labels for the neighbor and background instances. As mentioned earlier, the probability that the neighbor instances cover the object with the same category is smaller than the one of the PGT instances. The farther the spatial distance between the neighbor instance and the PGT instance is, the smaller the probability is. Based on the above-mentioned analysis, each neighbor instance is assigned a soft label, which is proportional to the spatial distance between the neighbor instance and its nearest PGT instance. Similarly, each background instance is also assigned a soft label which is inversely proportional to the spatial distance between the background instance and its nearest PGT instance.

The main contributions of this article are summarized as follows.

1) The PQS is proposed and adopted as the new metric to mine high-quality PGT instances for the training of multiple ICR branches. On the basis of traditional PCS, the DCPS which can measure the completeness that the proposal covers the object is proposed and incorporated into the PQS. The PGT instances mined by the PQS are more reliable and can better capture the whole object than the one solely relying on the PCS.
2) A novel PSLA strategy is proposed to assign more precise soft labels for the neighbor and background instances. The soft label of each neighbor (background) instance is (inversely) proportional to the spatial distance between the neighbor (background) instance and the nearest PGT instance. The proposed soft label is more precise than the traditional one-hot label, which can alleviate the misclassification problem.
3) The ablation study validates the effectiveness of the PQS and PSLA, and the comprehensive comparisons with other advanced WSOD methods show the excellent performance of our method on the Northwestern Polytechnical University very-high-resolution (NWPU VHR)-10.v2, detection in optical RSIs (DIOR), and PASCAL visual object classes (VOC) 2007 datasets.

II. Related Work

The WSDDN [21] and OICR [24] are two milestone works and have been adopted as basic architectures by our method and most WSOD methods; therefore, they are briefly introduced in this section.

A. Weakly Supervised Deep Detection Network

As shown in the left and top right parts of Fig. 2, first of all, a series of proposals $R = \{r_1, \ldots, r_n, \ldots, r_N\}$ are generated from input RSI by using selective search windows (SSWs) [29], where N indicates the number of proposals. Second, the proposal feature vectors of R are obtained via backbone network, region of interest (RoI) pooling, and two fully connected (FC) layers in turn. Third, the proposal feature vectors are separately imported into the classification and detection branch to obtain the classification and detection score matrix through two FC layers and softmax layer, and the elementwise product of two matrices is considered as the PCS matrix of R, which is denoted $\mathbf{x} \in \mathbb{R}^{C \times N}$. Fourth, the image-level prediction score ϕ_c of category c can be calculated by the following equation:

$$\phi_c = \sum_{n=1}^{N} x_{cn}, \quad c \in [1, C] \tag{1}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:06:51 UTC from IEEE Xplore. Restrictions apply.

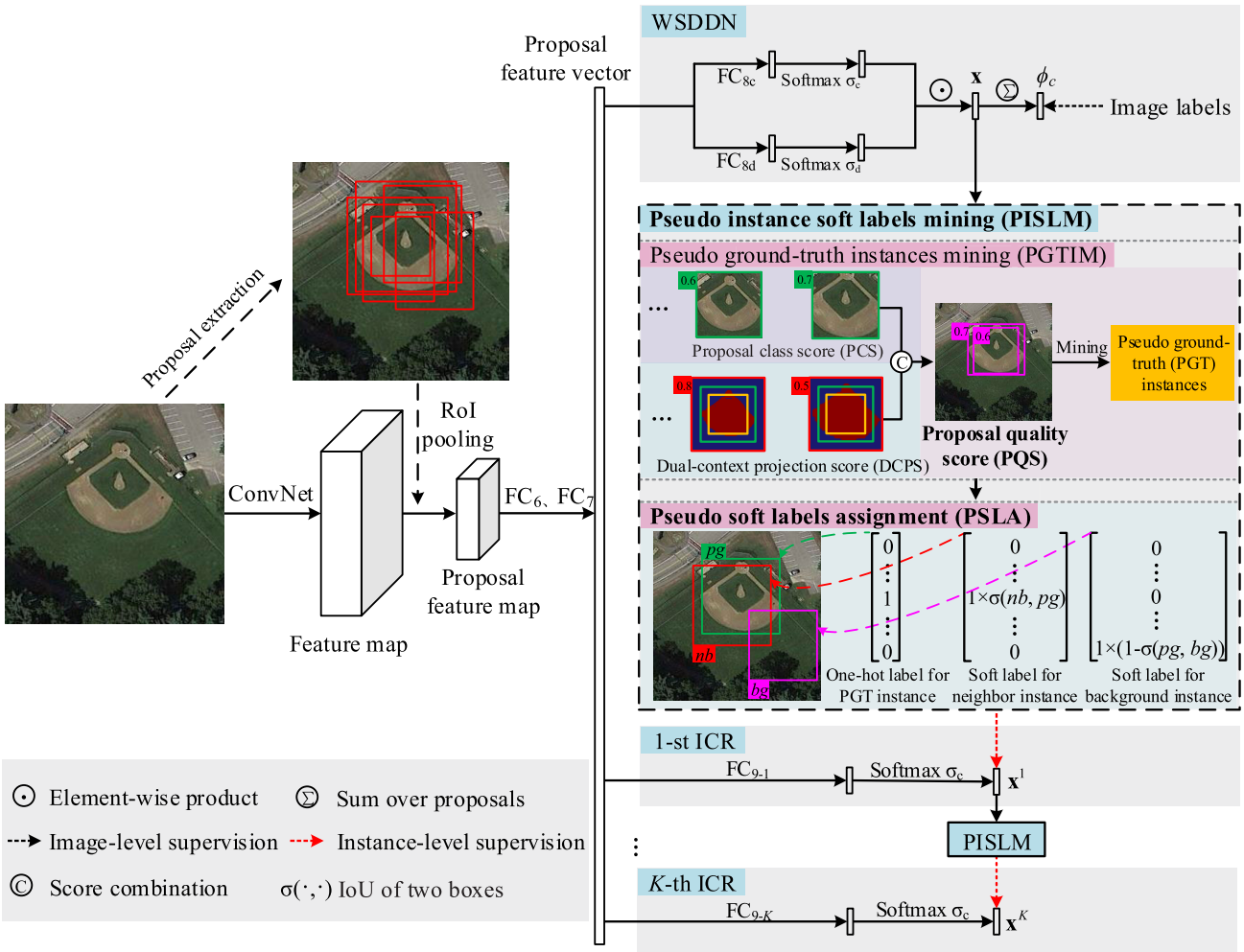


Fig. 2. Architecture of our method.

where C indicates the number of category, and $x_{cn} \in \mathbf{x}$ indicates the PCS that proposal r_n belongs to the cth category. In the end, the binary cross entropy is used to define the loss function L_w of WSDDN

$$L_w = -\sum_{c=1}^{C} (y_c \log \phi_c + (1 - y_c) \log(1 - \phi_c)) \tag{2}$$

where $y_c \in \{0, 1\}$ is the image-level label of input RSI, which denotes whether or not the input RSI contains the cth category objects.

B. Online Instance Classifier Refinement

The OICR incorporated K parallel ICR branches into WSDDN to further improve its performance. As shown in the lower right part of Fig. 2, each ICR branch contains an FC and a softmax layer which outputs a PCS matrix of R, denoted $\mathbf{x}^k \in \mathbb{R}^{(C+1)\times N}$, where $k \in 1, 2, \ldots, K$, $x_{cn}^k \in \mathbf{x}^k$ denotes the PCS that r_n belongs to the cth category, and the $(C+1)$th category denotes background. $\mathbf{x}^k$ is used to mine the pseudoinstance label matrix $\mathbf{y}^{k+1} \in \mathbb{R}^{(C+1)\times N}$ of the $(k+1)$th ICR branch, where $y_{cn}^{k+1} \in \mathbf{y}^{k+1}$ denotes whether or not the r_n belongs to the cth category. The $\mathbf{x}$ derived from WSDDN is used to mine the pseudoinstance label matrix $\mathbf{y}^1 \in \mathbb{R}^{(C+1)\times N}$ of the first ICR branch.

Finally, the kth ICR branch can be trained by the following loss function L_{ICR}^k:

$$L_{\text{ICR}}^k = -\frac{1}{N} \sum_{n=1}^{N} \sum_{c=1}^{C+1} w_n^k y_{cn}^k \log x_{cn}^k \tag{3}$$

where $w_n^k = x_{cn_c^{k-1}}^{k-1}$ denotes the loss weight of r_n, and the $x_{cn_c^{k-1}}^{k-1}$ denotes the highest PCS in the cth category for the $(k-1)$th ICR branch. $y_{cn}^k \in \{1, 0\}$ denotes whether or not the r_n belongs to the cth category in the kth ICR branch, its details are as follows. If $n = n_c^{k-1}$ or $r_n \in r_{n_c^{k-1}}^{nb}$, then $y_{cn}^k = 1$, otherwise, $y_{cn}^k = 0$, where $r_{n_c^{k-1}}$ denotes the proposal with the highest PCS, n_c^{k-1} is the index of $r_{n_c^{k-1}}$ and $r_{n_c^{k-1}}^{nb}$ denotes the assemble of neighbor proposals of $r_{n_c^{k-1}}$.

C. Other Related Works

The WSOD methods in natural scene images can provide lots of help to the research of WSOD methods in RSIs. Tang et al. [30] proposed a proposal cluster learning (PCL) strategy, which divided proposals into several clusters to mine multiple PGT instances. Wan et al. [31] proposed a min-entropy latent

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:06:51 UTC from IEEE Xplore. Restrictions apply.

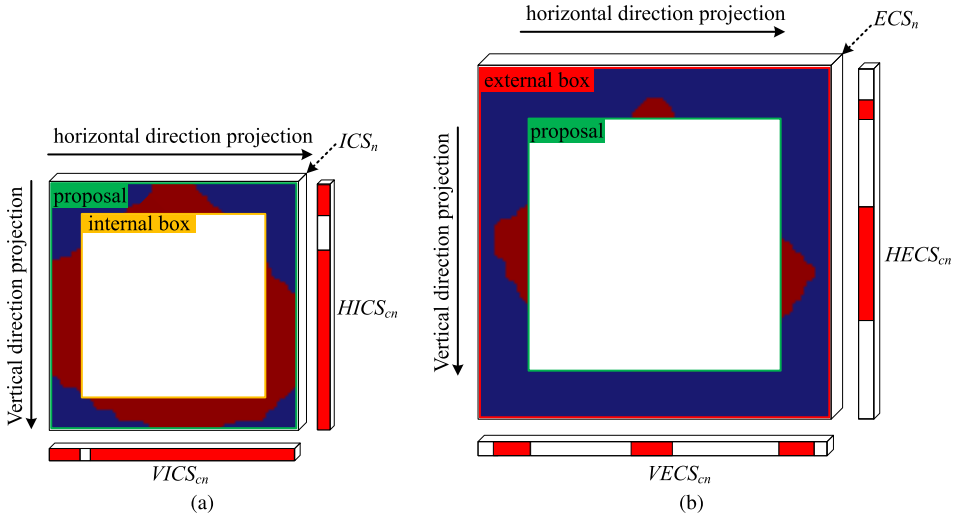


Fig. 3. Illustration of (a) ICPS and (b) ECPS.

model (MELM) to improve the discriminative capability of the WSOD model. Ren et al. [32] proposed a multiple instance self-training (MIST) scheme and a concrete drop block, which could mine more PGT instances and alleviate the part domination problem. Wu et al. [33] proposed an adaptive instance refinement (AIR) network to capture more instances with precise bounding boxes. Gao et al. [34] proposed a discrepant MIL (D-MIL) model, which was used to address the problems of instance missing and the localization of discriminative parts. Other classical works include coupled multiple instance detection network (C-MIDN) [35], proposal generation and proposal selection [36], object instance mining (OIM) [37], and so on.

The semantic segmentation can provide important cues to the WSOD model. For example, He et al. [38] proposed a transformer-induced hierarchical graph network (GraFNet) to improve the performance of semantic segmentation in RSIs. Peng and Ma [39] proposed two novel self-training strategies to mine more samples and alleviate the negative impact of wrong pseudolabels.

III. Proposed Method

A. Overview

As shown in Fig. 2, similar to many existing methods, the OICR is adopted as the basic framework of our method. The major difference between them is that a novel PISLM module is proposed to provide high-quality supervising information for K ICR branches. The PISLM module includes PGTIM and PSLA strategies. The PGTIM strategy introduces a novel metric named PQS and uses it to mine high-quality PGT instances. The PQS consists of the traditional PCS and the DCPS, where the DCPS is calculated through the weakly supervised semantic segmentation, and is employed to measure the completeness that each proposal covers an object. Instead of traditional one-hot labels, the more precise soft labels are assigned for the neighbor instances of PGT instances and background instances by the PSLA strategy. The soft label of each neighbor (background) instance is (inversely) proportional to the spatial distance between the neighbor (background) instance and the nearest PGT instance.

B. Pseudoground-Truth Instance Mining Guided by Proposal Quality Score

1) Proposal Quality Score: The proposed PGTIM strategy introduces the novel PQS to mine high-quality PGT instances, where the PQS is a combination of PCS and DCPS

$$\mathrm{PQS}_{cn}^{k} = \lambda \mathrm{PCS}_{cn}^{k} + (1-\lambda)\mathrm{DCPS}_{cn} \tag{4}$$

where the $\mathrm{PCS}_{cn}^{k} = x_{cn}^{k}$ denotes the PCS that r_n belongs to class c in the kth ICR branch, the DCPS_{cn} indicates the DCPS of r_n for class c, the PQS_{cn}^{k} denotes the quality score of r_n for class c in the kth ICR branch, and λ is a modulating factor for determining the relative weight of PCS_{cn}^{k} and DCPS_{cn}. The details of DCPS and λ are given as follows.

a) Dual-context projection score: The DCPS is designed to measure the completeness of that proposal covers the object. First of all, the weakly supervised semantic segmentation algorithm [40] is utilized to generate a semantic segmentation map of the input image. Second, the r_n is resized to $(1-\alpha)\times 100\%$ and $(1+\alpha)\times 100\%$ of the original size to obtain its internal box ir_n and external box er_n, as shown in Fig. 3, where $\alpha \in [0, 1]$ is a hyperparameter that is used to modulate the size of internal and external boxes. The r_n, ir_n, and er_n are then projected onto the semantic segmentation map of the input image to obtain their instance-level segmentation maps, denoted $S_n \in \mathbb{R}^{W\times H\times C}$, $IS_n \in \mathbb{R}^{IW\times IH\times C}$, and $\mathrm{ES}_n \in \mathbb{R}^{EW\times EH\times C}$, where W and H separately indicate the width and height of r_n, $IW = \lceil(1-\alpha)\times W\rceil$ and $IH = \lceil(1-\alpha)\times H\rceil$ separately indicate the width and height of the internal box, $EW = \lceil(1+\alpha)\times W\rceil$ and $EH = \lceil(1+\alpha)\times H\rceil$ separately indicate the width and height of the external box. Thirdly, the S_n and IS_n are employed to calculate the internal context projection score (ICPS) of r_n, while S_n and ES_n are employed to calculate the external context projection score (ECPS) of r_n.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:06:51 UTC from IEEE Xplore. Restrictions apply.

i) Calculation of ICPS: As shown in Fig. 3(a), the segmentation map $ICS_n \in \mathbb{R}^{W\times H\times C}$ of internal context region of r_n can be obtained through following equation:

$$\mathrm{ICS}_n = S_n - \mathrm{PIS}_n \tag{5}$$

where $\mathrm{PIS}_n \in \mathbb{R}^{W\times H\times C}$ denotes the zero-padding version of IS_n.

The projections of $\mathrm{ICS}_{cn} \in \mathbb{R}^{W\times H}$ (the cth channel of ICS_n) along the horizontal and vertical directions, denoted $\mathrm{HICS}_{cn} \in \mathbb{R}^{H}$ and $\mathrm{VICS}_{cn} \in \mathbb{R}^{W}$, can be calculated through the following equations:

$$\begin{aligned}\mathrm{HICS}_{cn} &= \mathrm{Max}_h(\mathrm{ICS}_{cn})\\ \mathrm{VICS}_{cn} &= \mathrm{Max}_v(\mathrm{ICS}_{cn})\end{aligned} \tag{6}$$

where $\mathrm{Max}_h(\cdot)$ and $\mathrm{Max}_v(\cdot)$ denote the maximizing operation along horizontal and vertical directions, respectively.

The ICPS of r_n for class c, denoted ICPS_{cn}, is obtained through the following equation:

$$\mathrm{ICPS}_{cn} = \mathrm{Avg}(\mathrm{HICS}_{cn}) + \mathrm{Avg}(\mathrm{VICS}_{cn}) \tag{7}$$

where $\mathrm{Avg}(\cdot)$ denotes averaging operation.

ii) Calculation of ECPS: As shown in Fig. 3(b), the segmentation map $\mathrm{ECS}_n \in \mathbb{R}^{EW\times EH\times C}$ of external context region of r_n can be obtained through the following equation:

$$\mathrm{ECS}_n = \mathrm{ES}_n - \mathrm{PS}_n \tag{8}$$

where $\mathrm{PS}_n \in \mathbb{R}^{EW\times EH\times C}$ denotes the zero-padding version of S_n.

The projections of $\mathrm{ECS}_{cn} \in \mathbb{R}^{EW\times EH}$ (the cth channel of ECS_n) along the horizontal and vertical directions, denoted $\mathrm{HECS}_{cn} \in \mathbb{R}^{EH}$ and $\mathrm{VECS}_{cn} \in \mathbb{R}^{EW}$, can be calculated through the following equations:

$$\begin{aligned}\mathrm{HECS}_{cn} &= \mathrm{Max}_h(\mathrm{ECS}_{cn})\\ \mathrm{VECS}_{cn} &= \mathrm{Max}_v(\mathrm{ECS}_{cn}).\end{aligned} \tag{9}$$

The ECPS of r_n for class c, denoted ECPS_{cn}, is obtained through the following equation:

$$\mathrm{ECPS}_{cn} = \mathrm{Avg}(\mathrm{HECS}_{cn}) + \mathrm{Avg}(\mathrm{VECS}_{cn}). \tag{10}$$

iii) Combining ICPS with ECPS for DCPS: The initial DCPS of proposal r_n for class c, denoted IDCPS_{cn}, can be obtained through the combination of ICPS_{cn} and ECPS_{cn}

$$\mathrm{IDCPS}_{cn} = \mathrm{ICPS}_{cn} - \mathrm{ECPS}_{cn}. \tag{11}$$

The motivation behind (11) is that a high-quality proposal should completely and tightly cover an object. Consequently, the proposal has rich internal context information and little external context information in this case. In other words, the IDCPS_{cn} is proportional to ICPS_{cn} and inversely proportional to ECPS_{cn}.

Finally, the IDCPS_{cn} are normalized to the range [0, 1] via the following linear transformation to obtain $\mathrm{DCPS}_{cn} \in [0, 1]$:

$$\mathrm{DCPS}_{cn} = \frac{\mathrm{IDCPS}_{cn} - \mathrm{Min}\{\mathrm{IDCPS}_{cn}\}_{n=1}^{N}}{\mathrm{Max}\{\mathrm{IDCPS}_{cn}\}_{n=1}^{N} - \mathrm{Min}\{\mathrm{IDCPS}_{cn}\}_{n=1}^{N}} \tag{12}$$

where $\mathrm{Max}\{\cdot\}$ and $\mathrm{Min}\{\cdot\}$ separately indicate the maximizing and minimizing functions. The motivation behind (12) is that the order of magnitude of DCPS is the same as the one of PCS, i.e., the value range of DCPS and PCS is [0, 1].

b) Modulating factor λ: As shown in (4), the λ is used to control the relative weight of PCS and DCPS. At the initial stage of training, the WSDDN and ICR branches have not been well trained, which induces that the reliability of PCS is not enough; therefore, the DCPS derived from the semantic segmentation map should be assigned a higher weight. As the training goes on, the reliability of PCS is continually improved, consequently, the λ should gradually transfer the focus from DCPS to PCS. Based on above-mentioned analysis, the λ is calculated through a growth function [33], which is formulated as follows:

$$\lambda = \frac{\log(t+v) - \log v}{\log(T+v) - \log v} \tag{13}$$

where t and T indicate the current and total numbers of iterations, respectively, and v is a hyperparameter for controlling the growth rate of λ.

2) Mining PGT Instances via Proposal Quality Score: Instead of traditional PCS_{cn}^{k-1}, the PQS_{cn}^{k-1} is used to mine the PGT instances of the kth ICR branch $\mathrm{PG}_c^k = \{pg_{c1}^k, \ldots, pg_{cm}^k, \ldots, pg_{cM}^k\}$ through the mining algorithm [32], where M denotes the number of PGT instances for class c in the kth ICR branch.

C. Pseudosoft Label Assignment

of Instances The existing methods usually assign a one-hot label for each instance in the kth ICR branch. As shown in Fig. 4(a), the PGT instance pg_{cm}^k is assigned a one-hot label vector in which the cth element is one. The neighbor instances of pg_{cm}^k, denoted NB_{cm}^k, are assigned the same label vectors as pg_{cm}^k. All of the background instances denoted BG^k, are assigned the same one-hot label vector in which the $(C+1)$th element is one. As mentioned earlier, the probability that the neighbor instances cover the objects with the same category is smaller than the PGT instances, the farther the spatial distance between the neighbor instance and the PGT instance is, the smaller the probability is. Therefore, the spatial distance factor should be involved in designing the label of NB_{cm}^k. Similarly, the spatial distance between background instances and their nearest PGT instances should also be considered in designing the label of BG^k. Based on the above-mentioned analysis, as shown in Fig. 4(b), a PSLA strategy is proposed for each instance, which is formulated as follows:

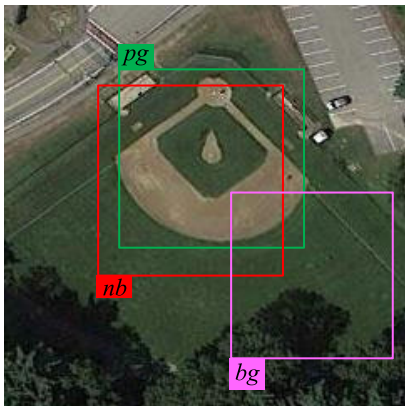
$$sy_{cn}^k = \begin{cases} 1, & \text{if } r_n \in \mathrm{PG}_c^k \\ 1 \times \sigma\left(r_n, pg_{cm}^k\right), & \text{if } r_n \in \mathrm{NB}_{cm}^k \\ 1 \times \left(1 - \sigma\left(r_n, nrpg_n^k\right)\right), & \text{if } r_n \in \mathrm{BG}^k \end{cases} \tag{14}$$

where $nrpg_n^k$ denotes the nearest PGT instance of r_n, sy_{cn}^k denotes the pseudosoft label of r_n for class c in the kth ICR branch, and $\sigma(\cdot,\cdot)$ denotes the intersection over union (IoU) of two boxes.

D. Overall Training Loss

Instead of the traditional one-hot label, the pseudosoft label of each instance is assigned according to (14); therefore, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:06:51 UTC from IEEE Xplore. Restrictions apply.



PGT instance | Neighbor instance | Background instance

$$\begin{bmatrix}0\\ \vdots\\ 1\\ \vdots\\ 0\end{bmatrix} \quad \begin{bmatrix}0\\ \vdots\\ 1\\ \vdots\\ 0\end{bmatrix} \quad \begin{bmatrix}0\\ \vdots\\ 0\\ \vdots\\ 1\end{bmatrix}$$

(a)

$$\begin{bmatrix}0\\ \vdots\\ 1\\ \vdots\\ 0\end{bmatrix} \quad \begin{bmatrix}0\\ \vdots\\ 1\times\sigma(nb,pg)\\ \vdots\\ 0\end{bmatrix} \quad \begin{bmatrix}0\\ \vdots\\ 0\\ \vdots\\ 1\times(1-\sigma(pg,bg))\end{bmatrix}$$

(b)

Fig. 4. Comparison between (a) traditional one-hot labels and (b) proposed pseudosoft labels.

training loss of the kth ICR branch defined in (3) is modified as follows:

$$L_{\text{ICR}}^{k} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{c=1}^{C+1} w_n^k s y_{cn}^k \log x_{cn}^k. \tag{15}$$

The overall training loss L of our method consists of two parts: the loss of WSDDN and K ICR branches

$$L = L_w + \sum_{k=1}^{K} L_{\text{ICR}}^{k}. \tag{16}$$

IV. Experiments

A. Experiment Setup

1) Dataset: Two popular RSI datasets (NWPU VHR-10.v2 [41], [42] and DIOR [43]) and one natural image dataset PASCAL VOC 2007 [44] are used to evaluate our method. The NWPU VHR-10.v2 dataset includes 1172 images with the size of 400 × 400, which contains ten categories and 2775 instances, and the training, validation, and testing subsets separately consist of 679, 200, and 293 images. The more challenging DIOR dataset includes 23 463 images with the size of 800 × 800, which contains 20 categories and 192 472 instances, and the training, validation, and testing subsets separately consist of 5862, 5863, and 11 738 images. The PASCAL VOC 2007 dataset includes 9963 images, which contains 20 categories and 24 640 instances, and the training, validation, and testing subsets separately consist of 2501, 2510, and 4952 images.

2) Evaluation Metric: The mean average precision (mAP) and correct localization (CorLoc) [45] are used for evaluating the overall detection accuracy and localization accuracy, respectively.

3) Implementation Details:

a) Algorithm setting: The OICR [24] is employed as the basic framework of our method for fair comparison [26], [46], [47]. The VGG-16 network [48] pretrained on the ImageNet dataset [10] is the backbone network. The threshold of non-maximum suppression (NMS) [49] is 0.3 [50].

b) Training setting: The stochastic gradient descent (SGD) algorithm is used for the optimization of our WSOD model. The initial learning rate, batch size, momentum, and weight decay are 0.01, 8, 0.9, and 0.0001, respectively. The top p percent and IoU threshold are set to 0.15 and 0.13, respectively. The total number of training iterations is 20K, 30K, and 60K on the NWPU VHR-10.v2, PASCAL VOC 2007, and DIOR datasets, respectively. This article adopts a step learning rate decay strategy with a decay weight of 0.1, and the step sizes are 12K and 16K iterations on the NWPU VHR-10.v2 dataset, 20K and 26K iterations on the PASCAL VOC 2007 dataset, and 50K and 56K iterations on the DIOR dataset, respectively. The training images are augmented via the horizontal flipping and rotation with 90° and 180°. The input images are resized to six scales {480, 576, 688, 864, 1000, 1200} for multiscale training and testing.

c) Environment setting: All experiments are conducted with PyTorch framework on eight GPUs (8 × 24-GB memory, Titan RTX).

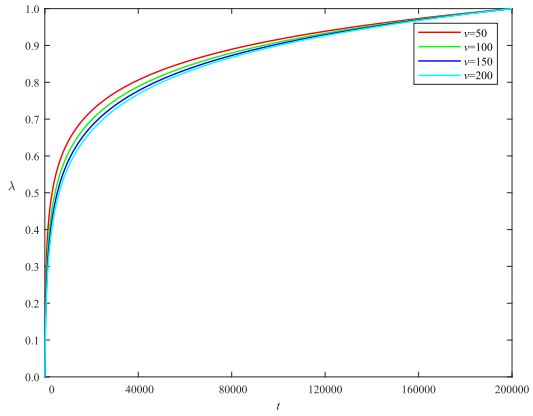
B. Parameter Analysis

1) Parameter Analysis of λ and v: As shown in Fig. 5(b), λ is a function of v (growth rate) and t (the number of iterations), and the function curves of λ about t with different v are shown in Fig. 5(a). As a matter of fact, λ can be considered as a symbiosis parameter of v. As shown in Fig. 5(b), the mAP and CorLoc of our method simultaneously obtain the highest value on the DIOR dataset when $v = 100$; therefore, the value of v is set to 100 in this article. λ can be solely determined by t when $v = 100$.

2) Parameter Analysis of α: As mentioned in Section III-B, α is used to modulate the size of internal and external boxes, and it is quantitatively analyzed on the DIOR dataset. As shown in Fig. 6, the mAP and CorLoc of our method simultaneously obtain the highest value when $\alpha = 0.2$; therefore, the value of α is set to 0.2 in this article.

3) Parameter Analysis of K: K (the number of ICR branches) is quantitatively analyzed on the DIOR dataset. As shown in Fig. 7, the mAP and CorLoc of our method simultaneously obtain the highest value when $K = 3$; therefore, the value of K is set to 3 in this article.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:06:51 UTC from IEEE Xplore. Restrictions apply.



(a)

λ	v	mAP	CorLoc
$\lambda = \frac{\log(t+v)-\log v}{\log(T+v)-\log v}$	50	26.3	51.2
	100	**26.9**	**52.4**
	150	26.4	51.6
	200	25.8	50.7

(b)

Fig. 5. Parameter analysis of λ and v on the DIOR dataset. (a) Function curves of λ about t with different v. (b) mAP and CorLoc values corresponding to different v (λ). T = 200K on the DIOR dataset.

C. Ablation Study

1) Quantitative Ablation Study: For validating the effectiveness of the proposed PSLA and PQS which consists of traditional PCS and the proposed DCPS, six methods with different configurations are compared with each other, as shown in Table I. Method A is the baseline method which only uses the traditional PCS. Method B is the combination of DCPS and the baseline method without PCS. Method C is the combination of DCPS and the baseline method, i.e., only PQS is adopted. Method D is the combination of PSLA and the baseline method. Method E is the combination of PSLA, DCPS, and the baseline method without PCS. Method F is the combination of PSLA, DCPS, and the baseline method.

As shown in Table I, the mAP and CorLoc of method B obtain the increase of 17.1% (7.0%) and 19.3% (14.4%) compared with baseline on the NWPU VHR-10.v2 (DIOR) datasets, which validates the effectiveness of the proposed DCPS and also indicates that the WSOD model based on the weakly supervised semantic segmentation results can give better performance than the traditional model.

The mAP and CorLoc of method C obtain the increase of 25.9% (10.4%) and 30.1% (17.6%) compared with the baseline on the NWPU VHR-10.v2 (DIOR) datasets, which validate the effectiveness of PQS. Furthermore, the performance of method C is superior to method B, which demonstrates that the combination of PCS and DCPS can obtain better performance. The reason for the accuracy increase by using PQS is as follows: the DCPS can quantitatively evaluate the completeness that each proposal covers the whole object; therefore, the PQS obtained by combining PCS with DCPS can alleviate the problem that the robustness is insufficient caused by solely using PCS, and can effectively mine high-quality PGT instances which covers the whole object.

The mAP and CorLoc of method D obtain the increase of 24.0% (9.2%) and 24.8% (16.6%) compared with the baseline on the NWPU VHR-10.v2 (DIOR) datasets, which validate the effectiveness of PSLA. The reason for the accuracy increase by using PSLA is as follows: the PSLA strategy can assign a more precise soft label for each instance according to the spatial distance between each instance and its nearest PGT instance, which alleviates the misclassification problem.

The mAP and CorLoc of method E obtain the increase of 25.1% (9.6%) and 27.3% (17.2%) compared with the baseline on the NWPU VHR-10.v2 (DIOR) datasets, which demonstrates that the combination of DCPS and PSLA is effective.

The mAP and CorLoc of our method are apparently higher than other methods on the two datasets, which demonstrates that the combination of PQS and PSLA is also effective.

2) Subjective Ablation Study: As shown in Fig. 8(a), the visual comparisons between baseline and method C on the NWPU VHR-10.v2 and DIOR datasets demonstrate that method C can capture more complete objects than baseline, which also verify the effectiveness of PQS. As shown in Fig. 8(b), the visual comparisons between baseline and method D on two datasets demonstrate that method D can reduce the false recognition of foreground objects compared with baseline, which also verifies the effectiveness of PSLA.

D. Comparison With Other Advanced WSOD Methods

For further validating the effectiveness of our method, it is quantitatively compared with 11 WSOD and seven FSOD methods on the NWPU VHR-10.v2 and DIOR datasets. The 11 WSOD methods include WSDDN [21], OICR [24], PCL [30], dynamic curriculum learning (DCL) [51], MIST [32], progressive contextual instance refinement (PCIR) [50], MELM [31], triple context-aware (TCA) [46], self-supervised adversarial and equivariant (SAE) [26], multiple instance graph (MIG) [47], and self-guided proposal generation (SPG) [25]. The five FSOD methods, including region-based convolutional neural network (R-CNN) [52], rotation-invariant convolutional neural networks (RICNN) [41], fast R-CNN [53], faster R-CNN [54] and detection-transformer with improved denoising anchor (DINO) [55], are used for comparisons on two RSI datasets. The two FSOD methods, including transferred CNN [10] and rotation-insensitive and context-augmented object detection (RICO) [42], are used for comparisons on the NWPU VHR-10.v2 dataset. The two FSOD methods, including RetinaNet [56] and path aggregation network (PANet) [57], are used for comparisons on the DIOR dataset.

Furthermore, to verify the generalization capacity of our method, it is quantitatively compared with ten WSOD and seven FSOD methods on the natural scene dataset PASCAL VOC 2007. The ten WSOD methods include WSDDN, OICR, PCL, MELM, OIM [37], PG-PS [36], C-MIDN [35], MIST, AIR [33], and D-MIL [34]. The seven FSOD methods include fast R-CNN, faster R-CNN, OHEM [58], R-FCN [59], PS-KD [60], CoupleNet [61], and DINO.

1) Comparison in Terms of mAP: As shown in Table II, the mAP of our method achieves 63.8% on the NWPU VHR-10.v2 dataset, which outperforms the WSDDN, OICR, PCL, MELM, DCL, MIST, PCIR, TCA, MIG, SAE, and SPG by 28.7%, 29.3%, 24.4%, 21.5%, 11.7%, 12.3%, 8.8%, 5.0%, 7.8%, 3.1%, and 1.0%, respectively. As shown in Table III, the mAP of our method achieves 28.6% on the more challenging

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:06:51 UTC from IEEE Xplore. Restrictions apply.

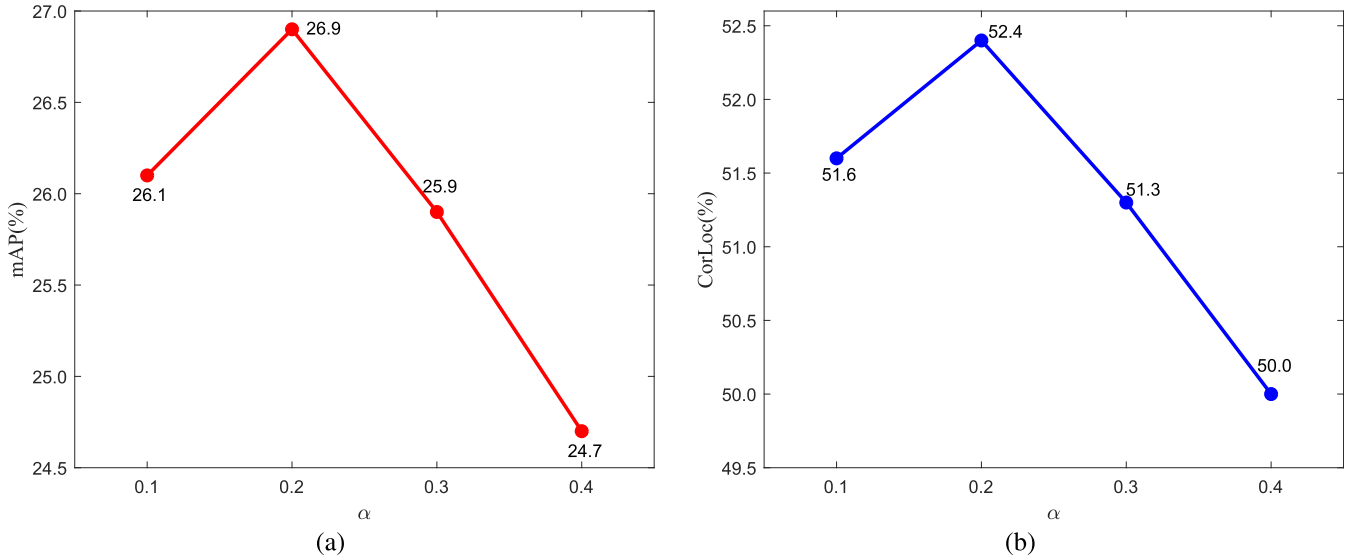


Fig. 6. Parameter analysis of α on the DIOR dataset. The values of (a) mAP and (b) CorLoc with respect to different α.

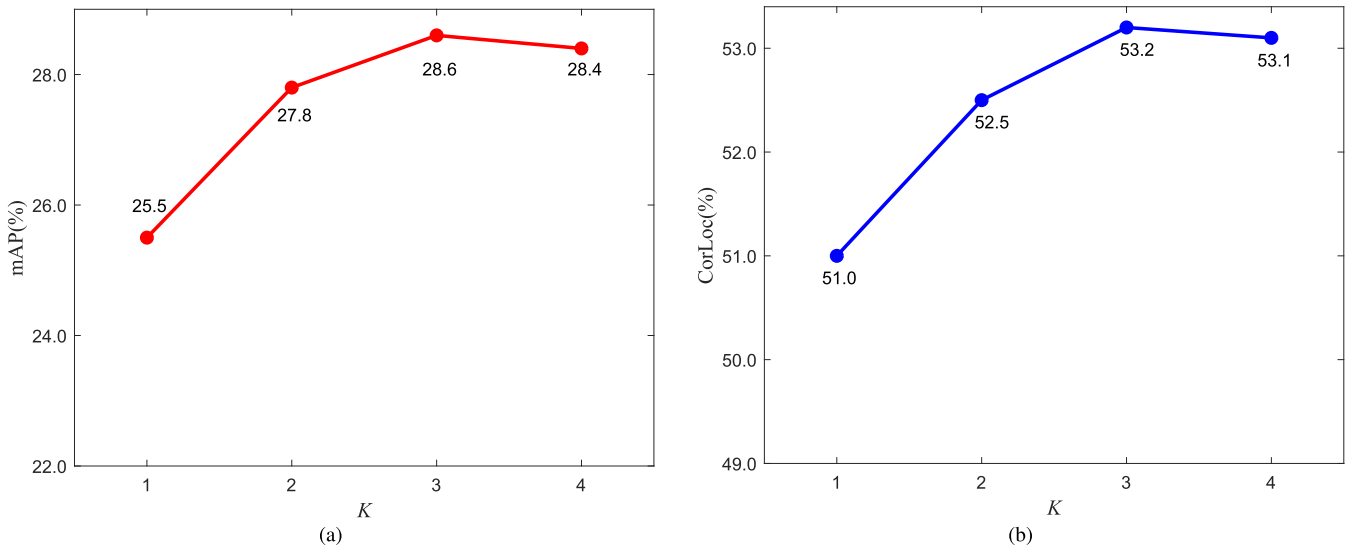


Fig. 7. Parameter analysis of K on the DIOR dataset. The values of (a) mAP and (b) CorLoc with respect to different K.

TABLE I

ABLATION STUDY ON THE NWPU VHR-10.V2 AND DIOR DATASETS. BOLD FONTS DENOTE THE BEST RESULTS. THE SAME BELOW

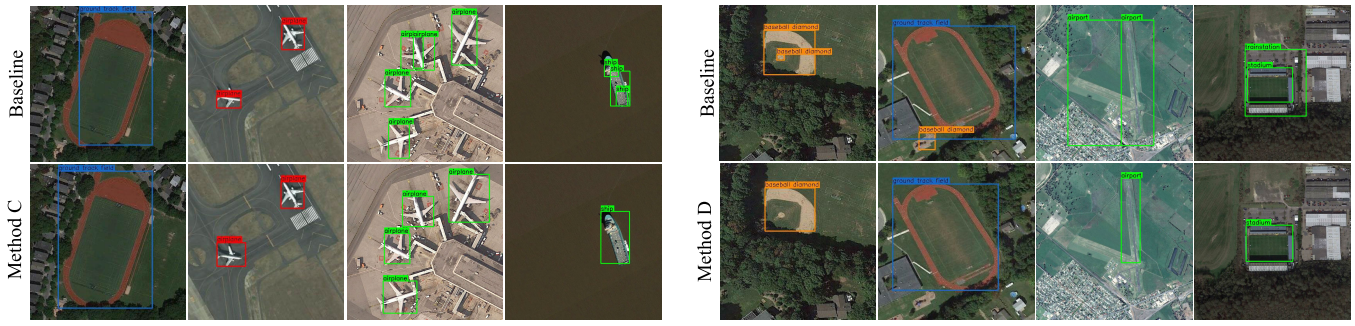
Method	PQS		PSLA	NWPU VHR-10.V2		DIOR	
	PCS	DCPS		mAP	CorLoc	mAP	CorLoc
A (Baseline)	✓			34.5	40.0	16.5	34.8
B		✓		51.6	59.3	23.5	49.2
C	✓	✓		60.4	70.1	26.9	52.4
D	✓		✓	58.5	64.8	25.7	51.4
E		✓	✓	59.6	67.3	26.1	52.0
F (Ours)	✓	✓	✓	**63.8**	**74.6**	**28.6**	**53.2**

DIOR dataset, which outperforms the WSDDN, OICR, PCL, MELM, DCL, MIST, PCIR, TCA, MIG, SAE, and SPG by 15.3%, 12.1%, 10.4%, 9.9%, 8.4%, 6.4%, 3.7%, 2.8%, 3.5%, 1.5%, and 2.8%, respectively. As shown in Table IV, the mAP of our method achieves 56.3% on the PASCAL VOC 2007 dataset, which outperforms the WSDDN, OICR, PCL, MELM, OIM, PG-PS, C-MIDN, MIST, AIR, and D-MIL by 21.5%, 15.1%, 12.8%, 9.0%, 6.2%, 5.2%, 3.7%, 1.4%, 2.8%, and 2.8%, respectively.

In a word, the performance of our method is superior to other advanced WSOD methods in terms of mAP, and the gap between WSOD and FSOD is narrowed by our method.

2) Comparison in Terms of CorLoc: As shown in Table V, the CorLoc of our method achieves 74.6% on the NWPU

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:06:51 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 8. Subjective evaluation of PQS and PSLA. (a) Comparisons between baseline and method C. (b) Comparisons between baseline and method D. The RSIs with green rectangles are selected from DIOR dataset, and the rest is selected from NWPU VHR-10.v2 dataset.

TABLE II
COMPARISONS WITH OTHER ADVANCED WSOD METHODS IN TERMS OF MAP (%) ON THE NWPU VHR-10.V2 DATASET

Method	Airplane	Ship	Storage tank	Baseball Diamond	Tennis Court	Basketball Court	Ground Track field	Harbor	Bridge	Vehicle	mAP
Transferred CNN [10]	66.0	57.1	85.0	80.9	35.1	45.5	79.4	62.6	43.2	41.3	59.6
R-CNN [52]	85.4	88.9	62.8	19.7	90.7	58.2	68.0	79.9	54.2	49.9	65.8
RICNN [41]	88.7	78.3	86.3	89.1	42.3	56.9	87.7	67.5	62.3	72.0	73.1
Fast R-CNN [53]	90.9	90.6	89.3	47.3	100.0	85.9	84.9	88.2	80.3	69.8	82.7
Faster R-CNN [54]	90.9	86.3	90.5	98.2	89.7	69.6	100.0	80.1	61.5	78.1	84.5
RICO [42]	99.7	90.8	90.6	92.9	90.3	80.1	90.8	80.3	68.5	87.1	87.1
DINO [55]	100.0	93.7	98.5	95.5	98.0	97.6	100.0	95.8	74.5	95.4	94.9
WSDDN [21]	30.1	41.7	35.0	88.9	12.9	23.9	99.4	13.9	1.9	3.6	35.1
OICR [24]	13.7	67.4	57.2	55.2	13.6	39.7	92.8	0.2	1.8	3.7	34.5
PCL [30]	26.0	63.8	2.5	89.8	64.5	76.1	77.9	0.0	1.3	15.7	39.4
MELM [31]	80.9	69.3	10.5	90.2	12.8	20.1	99.2	17.1	14.2	8.7	42.3
DCL [51]	72.7	74.3	37.1	82.6	36.9	42.3	84.0	39.6	16.8	35.0	52.1
MIST [32]	69.7	49.2	48.6	80.9	27.1	79.9	91.3	47.0	8.3	13.4	51.5
PCIR [50]	90.8	78.8	36.4	90.8	22.6	52.2	88.5	42.4	11.7	35.5	55.0
TCA [46]	89.4	78.2	78.4	90.8	35.3	50.4	90.9	42.4	4.1	28.3	58.8
MIG [47]	88.7	71.6	75.2	94.2	37.5	47.7	100.0	27.3	8.3	9.1	56.0
SAE [26]	82.9	74.5	50.2	96.8	55.7	72.9	100.0	36.5	6.3	31.9	60.7
SPG [25]	90.4	81.0	59.5	92.3	35.6	51.4	99.9	58.7	17.0	43.0	62.8
Ours	87.6	80.1	57.3	94.0	36.4	80.4	100.0	56.9	9.8	35.6	**63.8**

VHR-10.v2 dataset, which outperforms the WSDDN, OICR, PCL, MELM, MIST, PCIR, TCA, MIG, SAE, and SPG by 39.4%, 34.6%, 29.5%, 24.7%, 4.3%, 2.7%, 1.8%, 4.4%, 1.1%, and 1.2%, respectively. As shown in Table VI, the CorLoc of our method achieves 53.2% on the DIOR dataset, which outperforms the WSDDN, OICR, PCL, MELM, MIST, PCIR, TCA, MIG, SAE, and SPG by 20.8%, 18.4%, 11.7%, 9.9%, 9.6%, 7.1%, 4.8%, 6.4%, 3.8%, and 4.9%, respectively. As shown in Table VII, the CorLoc of our method achieves 70.3% on the PASCAL VOC 2007 dataset, which outperforms the WSDDN, OICR, PCL, MELM, OIM, PG-PS, C-MIDN, MIST, AIR, and D-MIL by 16.8%, 9.7%, 7.6%, 8.9%, 3.1%, 1.1%, 1.6%, 1.5%, 0.5%, and 1.6%, respectively.

In summary, the performance of our method outperforms other advanced WSOD methods in terms of CorLoc; furthermore, the relative superiority on the more challenging DIOR dataset is more obvious than on the NWPU VHR-10.v2 and PASCAL VOC 2007 datasets.

3) Subjective Comparison: To further evaluate our method, five WSOD methods that provide source codes are subjectively compared with our method on two RSI datasets. As shown in Fig. 9, our method can capture the foreground objects more completely, such as ground track field, and can alleviate the misclassification problem, such as ship.

E. Quantitative Evaluation for Various Degradations

To further verify the robustness of our method, it is quantitatively evaluated on the DIOR dataset when the RSIs suffer from various degradations, including noise, blurring, and compression. For the noise degradation, the Gaussian noise with a mean of 0 and variance of 100, the salt–pepper noise with a noise density of 0.01, and their combination are imposed on the original RSIs, respectively. For the blurring degradation, the Gaussian burring kernel with a mean of 0 and variance of 4 is imposed on the original RSIs. For the compression degradation, the jpeg compression with a jpeg quality of 80 is imposed on the original RSIs. The experimental results of original RSIs are considered as baseline.

Compared with the baseline, as shown in Table VIII, the mAP (CorLoc) of our method gets the decrease of 2.1%

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:06:51 UTC from IEEE Xplore. Restrictions apply.

TABLE III
COMPARISONS WITH OTHER ADVANCED WSOD METHODS IN TERMS OF MAP (%) ON THE DIOR DATASET

Method	Airplane	Airport	Baseball field	Basketball court	Bridge	Chimney	Dam	Expressway service area	Expressway toll station	Golf field
R-CNN [52]	35.6	43.0	53.8	62.3	15.6	53.7	33.7	50.2	33.5	50.1
RICNN [41]	39.1	61.0	60.1	66.3	25.3	63.3	41.1	51.7	36.6	55.9
Fast R-CNN [53]	44.2	66.8	67.0	60.5	15.6	72.3	52.0	65.9	44.8	72.1
Faster R-CNN [54]	50.3	62.6	66.0	80.9	28.8	68.2	47.3	58.5	48.1	60.4
RetinaNet [56]	53.7	77.3	69.0	81.3	44.1	72.3	62.5	76.2	66.0	77.7
PANet [57]	60.2	72.0	70.6	80.5	43.6	72.3	61.4	72.1	66.7	72.0
DINO [55]	87.7	91.7	86.9	90.3	56.5	82.9	75.3	91.7	81.1	84.1
WSDDN [21]	9.1	39.7	37.8	20.2	0.3	12.2	0.6	0.7	11.9	4.9
OICR [24]	8.7	28.3	44.1	18.2	1.3	20.2	0.1	0.7	29.9	13.8
PCL [30]	21.5	35.2	59.8	23.5	3.0	43.7	0.1	0.9	1.5	2.9
MELM [31]	28.1	3.2	62.5	28.7	0.1	62.5	0.2	28.4	13.1	15.2
DCL [51]	20.9	22.7	54.2	11.5	6.0	61.0	0.1	1.1	31.0	30.9
MIST [32]	32.0	39.9	62.7	29.0	7.5	12.9	0.3	5.1	17.4	51.0
PCIR [50]	30.4	36.1	54.2	26.6	9.1	58.6	0.2	9.7	36.2	32.6
TCA [46]	25.1	30.8	62.9	40.0	4.1	67.8	8.1	23.8	29.9	22.3
MIG [47]	22.2	52.6	62.8	25.8	8.5	67.4	0.7	8.9	28.7	57.3
SAE [26]	20.6	62.4	62.7	23.6	7.6	64.6	0.2	34.5	30.6	55.3
SPG [25]	31.3	36.7	62.8	29.1	6.1	62.7	0.3	15.0	30.1	35.0
Ours	29.1	49.8	70.9	41.4	7.2	45.5	0.2	35.4	36.8	60.8

Method	Ground track field	Harbor	Overpass	Ship	Stadium	Storage tank	Tennis court	Train station	Vehicle	Windmill	mAP
R-CNN [52]	49.3	39.5	30.9	9.1	60.8	18.0	54.0	36.1	9.1	16.4	37.7
RICNN [41]	58.9	43.5	39.0	9.1	61.1	19.1	63.5	46.1	11.4	31.5	44.2
Fast R-CNN [53]	62.9	46.2	38.0	32.1	71.0	35.0	58.3	37.9	19.2	38.1	50.0
Faster R-CNN [54]	67.0	43.9	46.9	58.5	52.4	42.4	79.5	48.0	34.8	65.4	55.5
RetinaNet [56]	74.2	50.7	59.6	71.2	69.3	44.8	81.3	54.2	45.1	83.4	65.7
PANet [57]	73.4	45.3	56.9	71.7	70.4	62.0	80.9	57.0	47.2	84.5	66.1
DINO [55]	85.3	66.7	66.7	79.9	82.0	78.3	91.9	67.6	63.4	93.0	80.2
WSDDN [21]	42.4	4.7	1.1	0.7	63.0	4.0	6.1	0.5	4.6	1.1	13.3
OICR [24]	57.4	10.7	11.1	9.1	59.3	7.1	0.7	0.1	9.1	0.4	16.5
PCL [30]	56.4	16.8	11.1	9.1	57.6	9.1	2.5	0.1	4.6	4.6	18.2
MELM [31]	41.1	26.1	0.4	9.1	8.6	15.0	20.6	9.8	0.0	0.5	18.7
DCL [51]	56.5	5.1	2.7	9.1	63.7	9.1	10.4	0.0	7.3	0.8	20.2
MIST [32]	49.5	5.4	12.2	29.4	35.5	25.4	0.8	4.6	22.2	0.8	22.2
PCIR [50]	58.5	8.6	21.6	12.1	64.3	9.1	13.6	0.3	9.1	7.5	24.9
TCA [46]	53.9	24.8	11.1	9.1	46.4	13.7	31.0	1.5	9.1	1.0	25.8
MIG [47]	47.7	23.8	0.8	6.4	54.1	13.2	4.1	14.8	0.2	2.4	25.1
SAE [26]	52.7	17.6	6.9	9.1	51.6	15.4	1.7	14.4	1.4	9.2	27.1
SPG [25]	48.0	27.1	12.0	10.0	60.0	15.1	21.0	9.9	3.2	0.1	25.8
Ours	48.5	14.0	25.1	18.5	48.9	11.7	11.9	3.5	11.3	1.7	**28.6**

TABLE IV
COMPARISONS WITH OTHER ADVANCED WSOD METHODS IN TERMS OF MAP (%) ON THE PASCAL VOC 2007

Method	Aero	Bike	Bird	Boat	Bottle	Bus	Car	Cat	Chair	Cow	Table	Dog	Horse	Mbike	Person	Plant	Sheep	Sofa	Train	Tv	mAP
Fast R-CNN [53]	74.5	78.3	69.2	53.2	36.6	77.3	78.2	82.0	40.7	72.7	67.9	79.6	79.2	73.0	69.0	30.1	65.4	70.2	75.8	65.8	66.9
Faster R-CNN [54]	70.0	80.6	70.1	57.3	49.9	78.2	80.4	82.0	52.2	75.3	67.2	80.3	79.8	75.0	76.3	39.1	68.3	67.3	81.1	67.6	69.9
OHEM [58]	77.7	81.9	76.0	64.9	55.8	86.3	86.0	86.8	53.2	82.9	70.3	85.0	86.3	78.7	78.0	46.8	76.1	72.7	80.9	75.5	75.1
R-FCN [59]	82.5	83.7	80.3	69.0	69.2	87.5	88.4	88.4	65.4	87.3	72.1	87.9	88.3	81.3	79.8	54.1	79.6	78.8	87.1	79.5	79.5
PS-KD [60]	83.5	86.0	79.2	72.7	65.1	86.7	88.2	88.9	64.1	86.8	72.0	87.1	87.7	84.0	81.3	56.1	78.1	79.8	87.9	79.2	79.7
CoupleNet [61]	85.7	87.0	84.8	75.5	73.3	88.8	89.2	89.6	69.8	87.5	76.1	88.9	89.0	87.2	86.2	59.1	83.6	83.4	87.6	80.7	82.7
DINO [55]	92.9	89.1	87.0	78.0	79.9	89.2	92.1	94.3	70.8	91.8	76.8	94.5	92.9	91.8	91.2	65.1	89.6	80.4	91.5	87.6	86.3
WSDDN [21]	39.4	50.1	31.5	16.3	12.6	64.5	42.8	42.6	10.1	35.7	24.9	38.2	34.4	55.6	9.4	14.7	30.2	40.7	54.7	46.9	34.8
OICR [24]	58.0	62.4	31.1	19.4	13.0	65.1	62.2	28.4	24.8	44.7	30.6	25.3	37.8	65.5	15.7	24.1	41.7	46.9	64.3	62.6	41.2
PCL [30]	54.4	69.0	39.3	19.2	15.7	62.9	64.4	30.0	25.1	52.5	44.4	19.6	39.3	67.7	17.8	22.9	46.6	57.5	58.6	63.0	43.5
MELM [31]	55.6	66.9	34.2	29.1	16.4	68.8	68.1	43.0	25.0	65.6	45.3	53.2	49.6	68.6	2.0	25.4	52.5	56.8	62.1	57.1	47.3
OIM [37]	55.6	67.0	45.8	27.9	21.1	69.0	68.3	70.5	21.3	60.2	40.3	54.5	56.5	70.1	12.5	25.0	52.9	55.2	65.0	63.7	50.1
PG-PS [36]	63.0	64.4	50.1	27.5	17.1	70.6	66.0	71.1	25.8	55.9	43.2	62.7	65.9	64.1	10.2	22.5	48.1	53.8	72.2	67.4	51.1
C-MIDN [35]	53.3	71.5	49.8	26.1	20.3	70.3	69.9	68.3	28.7	65.3	45.1	64.6	58.0	71.2	20.0	27.5	54.9	54.9	69.4	63.5	52.6
MIST [32]	68.8	77.7	57.0	27.7	28.9	69.1	74.5	67.0	32.1	73.2	48.1	45.2	54.4	73.7	35.0	29.3	64.1	53.8	65.3	65.2	54.9
AIR [33]	61.8	74.4	33.3	41.2	31.8	72.6	73.5	77.5	28.4	69.1	50.0	65.8	64.3	71.0	11.6	26.7	44.0	57.4	68.3	48.2	53.5
D-MIL [34]	60.4	71.3	51.1	25.4	23.8	70.4	70.3	71.9	25.2	63.4	42.6	67.1	57.7	70.1	15.5	26.6	58.7	63.3	66.9	67.6	53.5
Ours	67.3	69.1	55.9	37.0	31.3	68.4	77.2	70.7	31.1	69.3	42.2	61.6	60.1	76.5	28.8	26.4	57.9	60.3	67.2	68.2	**56.3**

(1.8%), 1.9% (1.5%), and 3.5% (3.6%) when RSI suffers from Gaussian noise, salt–pepper noise, and their combination, respectively. The mAP (CorLoc) of our method gets a decrease of 2.9% (3.1%) when RSI suffers from burring degradation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:06:51 UTC from IEEE Xplore. Restrictions apply.

TABLE V
COMPARISONS WITH OTHER ADVANCED WSOD METHODS IN TERMS OF CORLOC (%) ON THE NWPU VHR-10.V2 DATASET

Method	Airplane	Ship	Storage tank	Baseball Diamond	Tennis Court	Basketball Court	Ground Track field	Harbor	Bridge	Vehicle	CorLoc
WSDDN [21]	22.3	36.8	40.0	92.5	18.0	24.2	99.3	14.8	1.7	2.9	35.2
OICR [24]	29.4	83.3	20.5	81.8	40.9	32.1	86.6	7.4	3.7	14.4	40.0
PCL [30]	11.8	50.0	12.8	98.7	84.5	77.4	90.7	0.0	9.3	15.6	45.1
MELM [31]	86.0	77.4	21.4	98.3	10.7	43.5	95.0	40.0	11.8	14.6	49.9
MIST [32]	90.2	82.5	80.3	98.6	48.5	87.4	98.3	66.5	14.6	35.8	70.3
PCIR [50]	100.0	93.1	64.1	99.3	64.8	79.3	89.7	63.0	13.3	52.2	71.9
TCA [46]	96.9	91.8	95.1	88.7	66.9	62.8	96.0	54.2	19.6	55.6	72.8
MIG [47]	97.8	90.3	87.2	98.7	54.9	64.2	100.0	74.1	13.0	21.6	70.2
SAE [26]	97.1	91.7	87.8	98.7	40.9	81.1	100.0	70.4	14.8	52.2	73.5
SPG [25]	98.1	92.7	70.1	99.7	51.9	80.1	96.2	72.4	13.0	60.0	73.4
Ours	94.4	86.6	68.5	97.8	69.8	87.5	100.0	68.6	16.0	56.6	**74.6**

TABLE VI
COMPARISONS WITH OTHER ADVANCED WSOD METHODS IN TERMS OF CORLOC (%) ON THE DIOR DATASET

Method	Airplane	Airport	Baseball field	Basketball court	Bridge	Chimney	Dam	Expressway service area	Expressway toll station	Golf field
WSDDN [21]	5.7	59.9	94.2	55.9	4.9	23.4	1.0	6.8	44.5	12.8
OICR [24]	16.0	51.5	94.8	55.8	2.6	23.9	0.0	4.8	56.7	22.4
PCL [30]	61.1	46.9	95.4	63.6	7.3	95.1	0.2	5.7	5.1	50.8
MELM [31]	77.0	28.9	92.7	63.0	13.0	90.1	0.2	37.9	17.0	44.6
MIST [32]	91.6	53.2	93.5	66.3	10.8	30.7	1.5	14.3	35.2	47.5
PCIR [50]	93.1	45.6	95.5	68.3	3.6	92.1	0.2	5.4	58.4	47.5
TCA [46]	81.6	51.3	96.2	73.5	5.0	94.7	15.9	32.8	46.0	48.6
MIG [47]	77.0	46.9	95.4	63.6	23.0	95.1	0.2	17.0	57.9	50.8
SAE [26]	91.2	69.4	95.5	67.5	18.9	97.8	0.2	70.5	54.3	51.4
SPG [25]	80.5	32.0	98.7	65.0	15.2	96.1	22.5	17.0	46.1	51.0
Ours	85.5	68.9	96.8	75.8	11.6	94.7	0.8	67.5	60.5	46.5

Method	Ground track field	Harbor	Overpass	Ship	Stadium	Storage tank	Tennis court	Train station	Vehicle	Windmill	CorLoc
WSDDN [21]	89.9	5.5	10.0	23.0	98.5	79.6	15.1	3.5	11.6	3.2	32.4
OICR [24]	91.4	18.2	18.7	31.8	98.3	81.3	7.5	1.2	15.8	2.0	34.8
PCL [30]	89.4	42.1	19.8	37.9	97.9	80.7	13.8	0.2	10.5	6.9	41.5
MELM [31]	88.1	49.4	15.7	28.2	98.3	83.0	22.8	10.3	4.6	2.2	43.3
MIST [32]	87.1	38.6	23.4	50.7	80.5	89.2	22.4	11.5	22.2	2.4	43.6
PCIR [50]	88.6	15.8	5.2	39.5	98.1	85.6	13.4	56.5	9.7	0.6	46.1
TCA [46]	85.3	38.9	20.2	30.6	84.6	91.5	56.3	3.8	10.5	1.3	48.4
MIG [47]	89.4	42.1	19.8	37.9	97.9	80.7	13.8	10.3	10.5	6.9	46.8
SAE [26]	88.3	48.0	2.3	33.6	14.1	83.4	65.6	19.9	16.4	2.9	49.4
SPG [25]	89.2	49.5	22.0	35.2	98.6	90.0	32.6	12.7	10.0	2.3	48.3
Ours	75.2	50.5	28.3	39.7	92.6	77.0	55.1	10.1	20.9	5.6	**53.2**

TABLE VII
COMPARISON WITH OTHER ADVANCED WSOD METHODS IN TERMS OF CORLOC (%) ON THE PASCAL VOC 2007 DATASET

Method	Aero	Bike	Bird	Boat	Bottle	Bus	Car	Cat	Chair	Cow	Table	Dog	Horse	Mbike	Person	Plant	Sheep	Sofa	Train	Tv	CorLoc
WSDDN [21]	65.1	58.8	58.5	33.1	39.8	68.3	60.2	59.6	34.8	64.5	30.5	43.0	56.8	82.4	25.5	41.6	61.5	55.9	65.9	63.7	53.5
OICR [24]	81.7	80.4	48.7	49.5	32.8	81.7	85.4	40.1	40.6	79.5	35.7	33.7	60.5	88.8	21.8	57.9	76.3	59.9	75.3	81.4	60.6
PCL [30]	79.6	85.5	62.2	47.9	37.0	83.8	83.4	43.0	38.3	80.1	50.6	30.9	57.8	90.8	27.0	58.2	75.3	68.5	75.7	78.9	62.7
MELM [31]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	61.4
OIM [37]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	67.2
PG-PS [36]	85.4	80.4	69.1	58.0	35.9	82.7	86.7	82.6	45.5	84.9	44.1	80.2	84.0	89.2	12.3	55.7	79.4	63.4	82.1	82.1	69.2
C-MIDN [35]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	68.7
MIST [32]	87.5	82.4	76.0	58.0	44.7	82.2	87.5	71.2	49.1	81.5	51.7	53.3	71.4	92.8	38.2	52.8	79.4	61.0	78.3	76.0	68.8
AIR [33]	82.5	88.2	52.0	61.7	56.1	82.7	87.8	85.2	46.9	84.2	52.9	75.3	84.7	93.2	22.0	64.5	67.0	62.4	78.3	68.8	69.8
D-MIL [34]	81.3	82.0	72.7	48.9	42.0	80.2	86.1	78.5	43.9	80.2	42.2	76.5	68.7	91.2	32.7	56.0	81.4	69.6	78.7	79.9	68.7
Ours	80.6	85.6	70.8	54.1	49.6	80.7	88.0	76.0	45.1	87.4	48.4	70.2	76.1	90.6	36.1	59.4	81.8	70.1	73.0	83.1	**70.3**

The mAP (CorLoc) of our method gets a decrease of 1.5% (1.0%) when RSI suffers from compression degradation.

On the whole, the negative effect caused by various degradations is limited. As a matter of fact, the worst result (Gaussian

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:06:51 UTC from IEEE Xplore. Restrictions apply.



Fig. 9. Subjective comparisons between our method and other advanced WSOD methods on two datasets. The RSIs with green rectangles are selected from DIOR dataset, and the rest are selected from NWPU VHR-10.v2 dataset.

+ salt–pepper noise) is still in the top five (top two) compared with other advanced methods in terms of mAP (CorLoc), as shown in Tables III and VI.

F. Analysis of Some Failure Cases

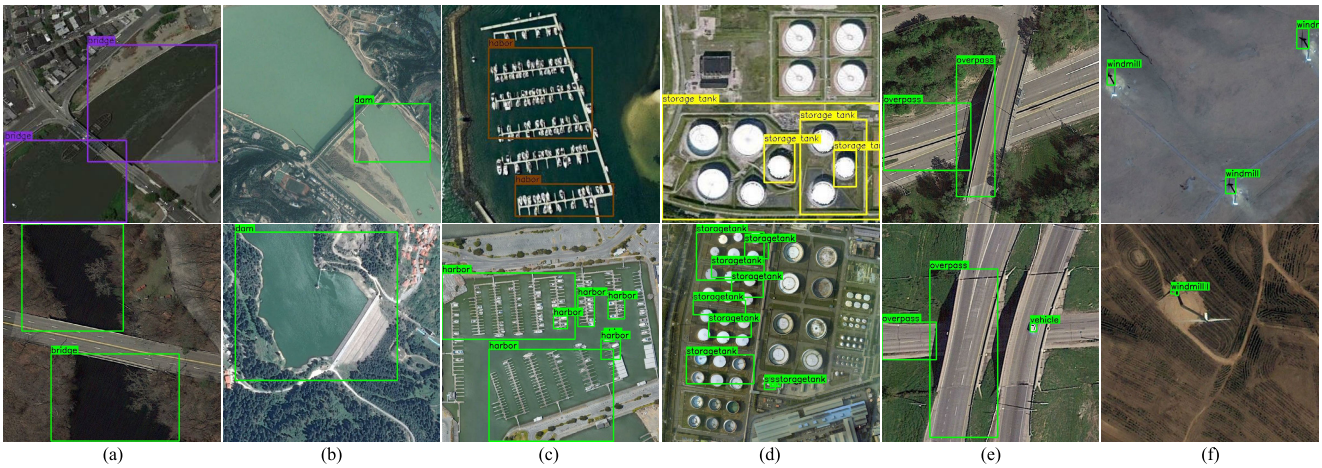
Despite our method having achieved better performance than other advanced WSOD methods, our method fails on individual classes, such as bridge, dam, harbor, and so on. The main reasons are as follows.

1) Coexisting Background: Some categories of RSIs must be determined by the core foreground objects and their coexisting background; therefore, the foreground objects and their coexisting background will be detected together by most of the WSOD methods without the instance-level annotations, and the same is true for our method. As shown

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:06:51 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
QUANTITATIVE EVALUATION OF OUR METHOD ON THE DIOR DATASET WHEN RSI SUFFERS FROM NOISE, BLURRING, AND COMPRESSION DEGRADATIONS

Metric	Original RSI (Baseline)	Noise			Blurring	Compression
		Gaussian noise	Salt-pepper noise	Gaussian+Salt-pepper noise		
mAP	28.6	26.5	26.7	25.1	25.7	27.1
CorLoc	53.2	51.4	51.7	49.6	50.1	52.2



(a) (b) (c) (d) (e) (f)

Fig. 10. Visualization of some failure results on two RSI datasets. The failures of (a) and (b) are induced by coexisting backgrounds, the failures of (c) and (d) are induced by dense objects, and the failures of (e) and (f) are induced by similar characteristics and imaging conditions, respectively. The RSIs with green rectangles are selected from DIOR dataset, and the rest are selected from NWPU VHR-10.v2 dataset.

in Fig. 10(a) and (b), the river and reservoir are coexisting background of the bridge and dam, respectively; therefore, they are detected together by our method.

2) Dense Objects: The overall features of multiple adjacent objects with the same category are usually more discriminative than the features of a single object when the objects are close to each other; therefore, multiple adjacent objects will be localized by a big bounding box. As shown in Fig. 10(c) and (d), the multiple adjacent harbors or storage tanks are captured within a big bounding box by our method.

3) Similar Characteristic: Some foreground objects have similar characteristics with other foreground objects or background regions; therefore, their detection will be disturbed. As shown in Fig. 10(e), the overpasses and roads have similar characteristics, so the roads are falsely detected as overpasses by our method.

4) Imaging Condition: The background regions are sometimes more salient than foreground objects because of various imaging conditions. As shown in Fig. 10(f), the shadows of windmills are more salient than windmills themself, so the shadows are falsely detected as windmills by our method.

V. CONCLUSION

A novel PISLM method is proposed to provide high-quality supervising information for WSOD in RSIs. The main contributions of our method include two parts: the PQS and PSLA strategy. The PQS consists of the traditional PCS and DCPS. The DCPS can measure the completeness that the proposal covers the object. The PQS is adopted as a novel metric to mine high-quality PGT instances, which is used to handle the problem that the PGT instances are mined by solely relying on PCS. As a matter of fact, the PGT instances mined by PCS are unreliable, and inclined to cover the most discriminative region, rather than the whole object. Instead of a traditional one-hot label, the PSLA strategy assigns a more precise soft label to each instance. The soft label is determined by the spatial distance between each instance and its nearest PGT instance and can alleviate the misclassification problem to some extent. The ablation study validates the effectiveness of the proposed PQS, PSLA, and their combination. The comprehensive comparisons with other advanced WSOD methods show the excellent performance of our method on the NWPU VHR-10.v2, DIOR, and PASCAL VOC 2007 datasets. It is worth noting that the proposed PISLM module can be considered a plug-and-play module and applied to other WSOD methods.

As mentioned in Section IV-F, the difficulty in object localization caused by the absence of instance-level annotations is the key reason for the failure of the WSOD model. To handle this situation, the bounding box regression branch can be added to the WSOD model. The pseudoinstance-level annotations can be mined by the weakly supervised semantic segmentation and are used to guide the training of the bounding box regression branch.

REFERENCES

[1] Y. Han, H. Liu, Y. Wang, and C. Liu, "A comprehensive review for typical applications based upon unmanned aerial vehicle platform," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9654–9666, 2022.

[2] W. Tang, C. Deng, Y. Han, Y. Huang, and B. Zhao, "SRARNet: A unified framework for joint superresolution and aircraft recognition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 327–336, 2021.

[3] Q. He, X. Sun, Z. Yan, B. Li, and K. Fu, "Multi-object tracking in satellite videos with graph-based multitask modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5619513.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:06:51 UTC from IEEE Xplore. Restrictions apply.

[4] Z. Wu, Y. Li, A. Plaza, J. Li, F. Xiao, and Z. Wei, "Parallel and distributed dimensionality reduction of hyperspectral data on cloud computing architectures," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 6, pp. 2270–2278, Jun. 2016.

[5] W. Tong, W. Chen, W. Han, X. Li, and L. Wang, "Channel-attention-based DenseNet network for remote sensing image scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4121–4132, 2020.

[6] G. Cheng, P. Lai, D. Gao, and J. Han, "Class attention network for image recognition," *Sci. China Inf. Sci.*, vol. 66, no. 3, Mar. 2023, Art. no. 132105.

[7] W. Chen, S. Ouyang, W. Tong, X. Li, X. Zheng, and L. Wang, "GCSANet: A global context spatial attention deep learning network for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1150–1162, 2022.

[8] J. Han et al., "Representing and retrieving video shots in human-centric brain imaging space," *IEEE Trans. Image Process.*, vol. 22, no. 7, pp. 2723–2736, Jul. 2013.

[9] X. Qian, Y. Zeng, W. Wang, and Q. Zhang, "Co-saliency detection guided by group weakly supervised learning," *IEEE Trans. Multimedia*, early access, Apr. 19, 2022, doi: 10.1109/TMM.2022.3167805.

[10] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Conf. Adv. Neural Inform. Process. Syst.*, 2012, pp. 1097–1105.

[11] X. Qian, J. Li, J. Cao, Y. Wu, and W. Wang, "Micro-cracks detection of solar cells surface via combining short-term and long-term deep features," *Neural Netw.*, vol. 127, pp. 132–140, Jul. 2020.

[12] J. Han, D. Zhang, X. Hu, L. Guo, J. Ren, and F. Wu, "Background prior-based salient object detection via deep reconstruction residual," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 25, no. 8, pp. 1309–1321, Aug. 2014.

[13] C. Deng, D. Jing, Y. Han, S. Wang, and H. Wang, "FAR-Net: Fast anchor refining for arbitrary-oriented object detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 6505805.

[14] X. Qian, S. Lin, G. Cheng, X. Yao, H. Ren, and W. Wang, "Object detection in remote sensing images based on improved bounding box regression and multi-level features fusion," *Remote Sens.*, vol. 12, no. 1, pp. 143–163, 2020.

[15] G. Cheng, C. Lang, M. Wu, X. Xie, X. Yao, and J. Han, "Feature enhancement network for object detection in optical remote sensing images," *J. Remote Sens.*, vol. 2021, pp. 1–10, Jan. 2021.

[16] X. Sun et al., "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 116–130, Feb. 2022.

[17] X. Qian, B. Wu, G. Cheng, X. Yao, W. Wang, and J. Han, "Building a bridge of bounding box regression between oriented and horizontal object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605209.

[18] G. Zhou, W. Chen, Q. Gui, X. Li, and L. Wang, "Split depth-wise separable graph-convolution network for road extraction in complex environments from high-resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5614115.

[19] Y. Yang, Z. Pan, Y. Hu, and C. Ding, "PistonNet: Object separating from background by attention for weakly supervised ship detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 5190–5202, Jun. 2022.

[20] D. Zhang, D. Meng, and J. Han, "Co-saliency detection via a self-paced multiple-instance learning framework," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 5, pp. 865–878, May 2016.

[21] H. Bilen and A. Vedaldi, "Weakly supervised deep detection networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2846–2854.

[22] A. Diba, V. Sharma, A. Pazandeh, H. Pirsiavash, and L. Van Gool, "Weakly supervised cascaded convolutional networks," in *Proc. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 914–922.

[23] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2018.

[24] P. Tang, X. Wang, X. Bai, and W. Liu, "Multiple instance detection network with online instance classifier refinement," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2843–2851.

[25] G. Cheng, X. Xie, W. Chen, X. Feng, X. Yao, and J. Han, "Self-guided proposal generation for weakly supervised object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5625311.

[26] X. Feng, X. Yao, G. Cheng, J. Han, and J. Han, "SAENet: Self-supervised adversarial and equivariant network for weakly supervised object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5610411.

[27] X. Qian et al., "Incorporating the completeness and difficulty of proposals into weakly supervised object detection in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1902–1911, 2022.

[28] X. Feng, X. Yao, G. Cheng, and J. Han, "Weakly supervised rotation-invariant aerial object detection network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14146–14155.

[29] J. R. R. Uijlings, K. E. A. van de Sande, T. Gevers, and A. W. M. Smeulders, "Selective search for object recognition," *Int. J. Comput. Vis.*, vol. 104, no. 2, pp. 154–171, Apr. 2013.

[30] P. Tang et al., "PCL: Proposal cluster learning for weakly supervised object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 1, pp. 176–191, Jan. 2020.

[31] F. Wan, P. Wei, J. Jiao, Z. Han, and Q. Ye, "Min-entropy latent model for weakly supervised object detection," in *Proc. IEEE CVPR*, Jun. 2018, pp. 1297–1306.

[32] Z. Ren et al., "Instance-aware, context-focused, and memory-efficient weakly supervised object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10598–10607.

[33] Z. Wu, J. Wen, Y. Xu, J. Yang, and D. Zhang, "Multiple instance detection networks with adaptive instance refinement," *IEEE Trans. Multimedia*, vol. 25, pp. 267–279, 2023.

[34] W. Gao, F. Wan, J. Yue, S. Xu, and Q. Ye, "Discrepant multiple instance learning for weakly supervised object detection," *Pattern Recognit.*, vol. 122, Feb. 2022, Art. no. 108233.

[35] G. Yan et al., "C-MIDN: Coupled multiple instance detection network with segmentation guidance for weakly supervised object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9834–9843.

[36] G. Cheng et al., "High-quality proposals for weakly supervised object detection," *IEEE Trans. Image Process.*, vol. 29, pp. 5794–5804, 2020.

[37] C. Lin, S. Wang, D. Xu, Y. Lu, and W. Zhang, "Object instance mining for weakly supervised object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 7, pp. 11482–11489.

[38] Q. He, X. Sun, W. Diao, Z. Yan, D. Yin, and K. Fu, "Transformer-induced graph reasoning for multimodal semantic segmentation in remote sensing," *ISPRS J. Photogramm. Remote Sens.*, vol. 193, pp. 90–103, Nov. 2022.

[39] C. Peng and J. Ma, "Domain adaptive semantic segmentation via entropy-ranking and uncertain learning-based self-training," *IEEE/CAA J. Autom. Sinica*, vol. 9, no. 8, pp. 1524–1527, Aug. 2022.

[40] J. Ahn, S. Cho, and S. Kwak, "Weakly supervised learning of instance segmentation with inter-pixel relations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 2209–2218.

[41] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[42] K. Li, G. Cheng, S. Bu, and X. You, "Rotation-insensitive and context-augmented object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2337–2348, Apr. 2017.

[43] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[44] M. Everingham, L. Van Gool, J. Winn, and A. Zisserman, "The PASCAL visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–308, Jun. 2010.

[45] T. Deselaers, B. Alexe, and V. Ferrari, "Weakly supervised localization and learning with generic knowledge," *Int. J. Comput. Vis.*, vol. 100, no. 3, pp. 275–293, Dec. 2012.

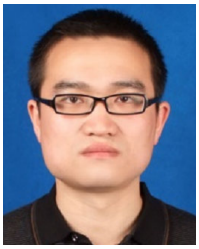
[46] X. Feng, J. Han, X. Yao, and G. Cheng, "TCANet: Triple context-aware network for weakly supervised object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6946–6955, Aug. 2021.

[47] B. Wang, Y. Zhao, and X. Li, "Multiple instance graph learning for weakly supervised remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5613112.

[48] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Represent.*, 2015, pp. 1–13.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:06:51 UTC from IEEE Xplore. Restrictions apply.

[49] J. Hosang, R. Benenson, and B. Schiele, "Learning non-maximum suppression," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 4507–4515.
[50] X. Feng, J. Han, X. Yao, and G. Cheng, "Progressive contextual instance refinement for weakly supervised object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8002–8012, Nov. 2020.
[51] X. Yao, X. Feng, J. Han, G. Cheng, and L. Guo, "Automatic weakly supervised object detection from high spatial resolution remote sensing images via dynamic curriculum learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 675–685, Jan. 2021.
[52] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 580–587.
[53] R. Girshick, "Fast R-CNN," in *Proc. ICCV*, Sep. 2015, pp. 1440–1448.
[54] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2016.
[55] H. Zhang et al., "DINO: DETR with improved DeNoising anchor boxes for end-to-end object detection," 2022, *arXiv:2203.03605*.
[56] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.
[57] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.
[58] A. Shrivastava, A. Gupta, and R. Girshick, "Training region-based object detectors with online hard example mining," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 761–769.
[59] J. Dai, Y. Li, K. He, and J. Sun, "R-FCN: Object detection via region-based fully convolutional networks," in *Proc. Conf. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 1–9.
[60] K. Kim, B. Ji, D. Yoon, and S. Hwang, "Self-knowledge distillation with progressive refinement of targets," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 6567–6576.
[61] Y. Zhu, C. Zhao, J. Wang, X. Zhao, Y. Wu, and H. Lu, "CoupleNet: Coupling global structure with local parts for object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4126–4134.



Xiaoliang Qian (Member, IEEE) received the Ph.D. degree from the School of Automation, Northwestern Polytechnical University, Xi'an, China, in 2013.

He is currently an Associate Professor with the School of Electrical and Information Engineering, Zhengzhou University of Light Industry, Zhengzhou, China. His research interests include remote sensing image understanding, deep learning, and computer vision.



Yu Huo received the B.S. and M.S. degrees from the School of Electrical and Information Engineering, Zhengzhou University of Light Industry, Zhengzhou, China, in 2019 and 2022, respectively.

He is currently a Research Assistant with the School of Electrical and Information Engineering, Zhengzhou University of Light Industry. His research interests include object detection in remote sensing images and deep learning.



Gong Cheng (Member, IEEE) received the B.S. degree from Xidian University, Xi'an, China, in 2007, and the M.S. and Ph.D. degrees from Northwestern Polytechnical University, Xi'an, in 2010 and 2013, respectively.

He is currently a Professor with Northwestern Polytechnical University. His research interests include computer vision, pattern recognition, and remote sensing image understanding.

Dr. Cheng is also an Associate Editor of the IEEE GEOSCIENCE AND REMOTE SENSING MAGAZINE and a Guest Editor of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.



Chenyang Gao received the B.E. degree from Nanchang University, Nanchang, Jiangxi, China, in 2020. She is currently pursuing the M.S. degree with Northwestern Polytechnical University, Xi'an, China.

Her research interests include computer vision and remote sensing image processing, with an emphasis on object detection.



Xiwen Yao (Member, IEEE) received the B.S. and Ph.D. degrees from Northwestern Polytechnical University, Xi'an, China, in 2010 and 2016, respectively.

He is currently an Associate Professor with Northwestern Polytechnical University. His research interests include computer vision and remote sensing image processing, with an emphasis on fine-grained image classification and object detection.



Wei Wang received the Ph.D. degree from Concordia University, Montréal, QC, Canada, in 2002.

He is currently a Professor with the School of Electrical and Information Engineering, Zhengzhou University of Light Industry, Zhengzhou, China. His research interests include artificial intelligence, remote sensing image understanding, computer vision, and tactile sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:06:51 UTC from IEEE Xplore. Restrictions apply.